# Use Case Descriptions

MTA Daily Ridership: Post-COVID Recovery Analysis & Next-Month Forecast | Phase 2: Design & Analysis | October 2026

This document supports **03_UseCase_Diagram**. It lists the actors, every use case with its linked user story and functional requirement, and the full description of the most important use cases. User stories (US) and requirements (FR) come from **04_Requirements** (SRS and Traceability Matrix).

## 1. Actors

| Actor (in diagram) | Stakeholder in SRS | What the actor needs from the dashboard |
|---|---|---|
| MTA Executive Board | MTA Executive Board | High-level recovery trend, KPI scorecard, YoY growth and the November 2024 forecast. |
| Transit Service Planner | Transit Service Planners | Mode-by-mode recovery against pre-pandemic levels, seasonal peaks, transparent handling of data anomalies. |
| Operations Manager | Operations Managers | Weekday vs. Weekend patterns and seasonality for staff scheduling. |
| Data & BI Analyst | Data & BI Team (Authors) | Clean data, passenger vs. non-passenger separation, filters; also uses all dashboard pages. |
| DEPI Evaluator | DEPI Evaluation Committee | Traceability, documentation and version control (README, ETL steps, DAX dictionary). |

Commuters / Public Representatives are low-influence beneficiaries and do not use the dashboard directly, so they are not an actor.

## 2. Use case summary

| ID | Use case | Main actor(s) | Relationship | User story | FR |
|---|---|---|---|---|---|
| UC-01 | View KPI Scorecard | Executive | Extended by UC-09 | US-09 | FR-02, FR-07, FR-11 |
| UC-02 | Filter by Date and Mode | Analyst (used by all) | Included by UC-03 to UC-07 | US-08 | FR-06, FR-11 |
| UC-03 | Track Ridership Trend | Executive | includes UC-02 | US-01 | FR-01, FR-02, FR-06, FR-09 |
| UC-04 | Analyze Recovery vs Pre-Pandemic | Planner (Executive also views) | includes UC-02 | US-02 | FR-03 |
| UC-05 | Compare Weekday vs Weekend | Operations Manager | includes UC-02 | US-05 | FR-05 |
| UC-06 | Identify Seasonal Peaks & Top 3 Days | Planner, Operations Manager | includes UC-02 | US-11 | FR-10 |
| UC-07 | View Next-Month Forecast | Executive | includes UC-02 and UC-08 | US-03 | FR-02, FR-04 |
| UC-08 | View Forecast Accuracy (MAPE) | (through UC-07) | Included by UC-07 | US-03 | FR-04 |
| UC-09 | View YoY Growth | Executive | extends UC-01 (optional card) | US-06 | FR-05 |
| UC-10 | Clean Data & Handle Anomalies | Analyst (benefits Planner) | - | US-07 | FR-01, FR-08 |
| UC-11 | Separate Passenger from Non-Passenger Metrics | Analyst | - | US-04 | FR-02 |
| UC-12 | Review Documentation & Lineage | DEPI Evaluator | - | US-10 | FR-08 |

Coverage: all 11 user stories (US-01 to US-11) are linked to at least one use case.

## 3. UC-02 Filter by Date and Mode

| | |
|---|---|
| **Actors** | Data & BI Analyst (and every actor, through the use cases that include it) |
| **Pre-condition** | The dashboard is open and the data model is loaded. |
| **Main flow** | 1) The user opens the slicer pane or the slicers on the page. 2) The user selects a Year/Month and/or a Transit Mode. 3) The system filters every visual on the page and keeps the selection across pages. |
| **Post-condition** | All visuals show only the selected period and mode. |
| **Quality target** | Visuals refresh in under 3 seconds (NFR-01). |
| **Linked** | US-08 \| FR-06, FR-11 |

## 4. UC-04 Analyze Recovery vs Pre-Pandemic

| | |
|---|---|
| **Actors** | Transit Service Planner (Executive and Analyst can also view) |
| **Pre-condition** | The dashboard is open and data is loaded. |
| **Main flow** | 1) The user opens the Recovery page. 2) The user selects a mode and a period (UC-02). 3) The system shows the % of comparable pre-pandemic day for each of the 7 modes against a static 100% reference line. |
| **Post-condition** | The user can identify which modes have recovered and which are lagging. |
| **Linked** | US-02 \| FR-03 |

## 5. UC-05 Compare Weekday vs Weekend

| | |
|---|---|
| **Actors** | Operations Manager |
| **Pre-condition** | The dashboard is open and data is loaded. |
| **Main flow** | 1) The user opens the Patterns page. 2) The user selects a Transit Mode (UC-02). 3) The system shows average ridership for Weekday and Weekend and the ratio between them. |
| **Post-condition** | The user can see the weekday/weekend gap for the selected mode and plan staffing. |
| **Linked** | US-05 \| FR-05 |

## 6. UC-06 Identify Seasonal Peaks & Top 3 Days

| | |
|---|---|
| **Actors** | Transit Service Planner, Operations Manager |
| **Pre-condition** | The dashboard is open and data is loaded. |
| **Main flow** | 1) The user opens the Patterns page. 2) The user selects a mode and period (UC-02). 3) The system shows average passenger ridership by calendar month and lists the top 3 peak days. |
| **Post-condition** | The user can name the highest and lowest months and the 3 busiest days. |
| **Linked** | US-11 \| FR-10 |

## 7. UC-07 View Next-Month Forecast

| | |
|---|---|
| **Actors** | MTA Executive Board |
| **Pre-condition** | The dashboard is open and the forecast values are calculated. |
| **Main flow** | 1) The user opens the Forecast page. 2) The user selects a period (UC-02). 3) The system shows the forecast of total passenger ridership for November 2024 (sum of daily forecasts) against actuals. 4) The system also shows the MAPE of the backtest (UC-08). |
| **Post-condition** | The user has the November 2024 figure and its measured accuracy (target: MAPE of 10% or less). |
| **Linked** | US-03 \| FR-02, FR-04 |

## 8. Other use cases

| ID | Short description |
|---|---|
| UC-01 | The Executive opens the Overview page and sees the KPI cards: Total Passenger Ridership, Overall Recovery %, November Forecast. UC-09 adds the YoY growth card. |
| UC-03 | The Executive views daily and monthly total passenger ridership since March 2020 and narrows the period with Year/Month slicers. |

| ID | Short description |
|---|---|
| UC-08 | Always shown with the forecast: the MAPE card from the backtest on August to October 2024. If MAPE is above the target, the value is still shown and the gap is documented. |
| UC-09 | Optional card with YoY growth %, calculated with DAX Time Intelligence. |
| UC-10 | The Analyst reviews zero values in Staten Island Railway and converts confirmed missing observations to nulls in Power Query, then logs the decision. |
| UC-11 | The Analyst keeps Total Passengers to Subways, Buses, LIRR, Metro-North and Staten Island Railway; Bridges & Tunnels and Access-A-Ride use separate measures. |
| UC-12 | The Evaluator opens the GitHub repository and checks the README (Power Query steps, DAX dictionary) and version history. |